GENERALIZED STATE ESTIMATION

O. Alsaç, N. Vempati, B. Stott
PCA Corporation
Mesa, Arizona, USA

A. Monticelli
UNICAMP
Campinas, SP, Brazil

Abstract

Power system state estimation derives a real-time network model by extracting information from a redundant data set consisting of telemetered, predicted and static data items. This paper describes a generalized, fully developed, estimation approach that fundamentally improves the information extraction process. Its main contribution is the successful inclusion of topology and parameters in the estimation and bad data analysis processes. This is valuable both in the initial commissioning of a state estimator, and in its routine real-time and study mode application. The approach involves a variety of novel concepts and methods. It is usable in Weighted Least Squares (WLS) and other estimation approaches.

Keywords: *State Estimation, Topology Processing, Observability Analysis, Bad Data Analysis, Topology Errors, Parameter Estimation*

INTRODUCTION

Power system state estimation produces a real-time "snapshot" model of the network based on three main classes of data: analog measurements, switch (breaker) statuses and device parameters. The conventional WLS approach [1] minimizes a scalar $R^t.W.R$, where element i of measurement residual vector R is:

$$r_i = z_i - h_i(X) \tag{1}$$

Here, z_i is the analog measurement and h_i is its analytical description. State vector X (the complex bus voltages and possibly transformer taps) is calculated by recursively forming the Jacobian matrix $H = \partial h / \partial x$ and solving the gain matrix equation:

$$G.\Delta X = H^t.W.H.\Delta X = H^t.W.R \tag{2}$$

Simplifying Assumptions in Conventional Estimation

Conventional estimation employs two critical simplifications: the interactions between the three data classes are decoupled, and the bus-oriented (system planning) reduction of the physical network model is adopted. This permits the familiar estimation sequence:

1. Perform network topology processing,
   - ***assuming that the switch statuses are correct.***
2. Construct the network bus-oriented model,
   - ***assuming sufficiently accurate network parameters.***
3. Use the results of 1-2 to find the observable islands.
4. Use the results of 1-3 to estimate the network state.
5. Use the results of 1-4 to filter out any analog bad data.

PE-186-PWRS-16-09-1997 A paper recommended and approved by the IEEE Power System Engineering Committee of the IEEE Power Engineering Society for publication in the IEEE Transactions on Power Systems. Manuscript submitted May 27, 1997; made available for printing September 30, 1997.

The foregoing switch and parameter assumptions are fundamentally flawed. The conventional estimation approach is efficient and works well only when these assumptions happen to be valid, and otherwise can severely malfunction. To illustrate, a topology error typically appears in the guise of multiple interacting bad data. Bad data analysis methods then tend to rate neighboring good analog measurements as gross errors, and eliminate them from the measurement set. As soon as local redundancy is lost, the indications of bad data disappear. But the topology error remains.

Sophisticated techniques such as topology pre-processing, local post-solution parameter estimation, expert systems, etc. only provide partial recovery from the incorrect assumptions. Formulations that avoid these assumptions in the first place seem to be needed.

A Generalized Approach

The generalized approach of this paper formulates and analyzes the analog, switch and impedance data **as a single interacting information set**. In consequence, some portions of the network become modeled at bus-section/switch (breaker) level, and designated network parameters become estimated quantities. The existing set of measurements is extended, and eq.(1) becomes:

$$r_i = z_i - h_i(X, S, Y) \tag{3}$$

where the voltages of modeled bus sections are added to X, S represents the flows through modeled switches [2], and Y represents the flows through network impedances of uncertain value.

Translating the generalized approach into practice requires:

- Rethinking some of the interactions between the data classes and their errors. This has novel and far-reaching consequences. Now that observability, topology and bad data analyses are all interdependent, parts of the network have to be solved that would normally be ignored as de-energized and unobservable.
- New top-down design and code, to handle non-traditional devices and measurements, to accommodate arbitrary mixes of the bus/branch and bus-section/switch models, and to automatically swap any portions of the network between these models.

One of the big challenges arises from the recognition that in future, state estimation will be performed on networks of increasing size, at shorter time intervals, and probably with dynamic extensions. For the sake of efficiency we therefore sought the flexibility to:

- Invoke the power of the generalized approach only as needed. It should have the fast conventional estimation as a subset.
- Exploit localization in bad data analysis as much as possible.

The present generalized approach obviously draws on a range of previous methods and ideas [2-7]. Here, multiple ideas have been refined, fully worked out and integrated: the generalized estimator is already being installed in utility control centers. The paper describes the approach in as much detail as space permits.

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MODELING FOR GENERALIZED APPROACH

Switch Modeling

Conventional state estimation is performed on a reduced network, in which the bus sections and switches of the physical system have been eliminated, assuming correct switch statuses. In contrast, a non-reduced physical network model has various attractions [2-4]:

- Measurements on switches and other zero or low impedance branches are trivially included.
- Low impedance branches, such as short cables with large shunt susceptances, are handled.
- The flow in each switch and the voltage at each bus section can now be calculated, subject to observability.
- *The explicit modeling of switches enormously facilitates switch status bad data analysis.*

A consequence of the above is that no local equivalencing or other manual data "massaging" for state estimation are needed. The data base maintains a clean consistent description of the actual system.

As shown in Table 1, each switch is automatically modeled by:

- Making its power throughflow P+jQ a state variable.
- Adding as a pseudo-measurement: (a) zero-flow for an "open" switch, (b) zero-voltage-drop for a "closed" switch. **This pseudo-measurement is subject to bad data analysis.** An "unknown" switch gets no pseudo-measurement. Series or parallel switches in paths with ambiguous statuses are automatically combined into "logical switches".

Each bus section is automatically modeled like a bus:

- Its voltage V,θ becomes a state variable.
- A zero-injection pseudo-measurement is added when it has no actual injection.

Modeling of Network Parameters

Each series or shunt parameter to be estimated is removed from the network model, and is replaced by state variables representing its terminal power flows (see Table 1). As in the case of switches and zero impedance branches, this avoids the explicit use of branch parameters during the estimation process. Once the solved states have been obtained, the device impedances are trivially calculated.

Also:

- In the case of a series impedance, a pseudo-measurement requiring the current flows at each end to be the same is automatically introduced.
- In the case of a shunt admittance of a symmetrical-pi transmission line, a pseudo-measurement requiring the legs at each end to be the same is automatically introduced.

Naturally, parameter estimation can only be applied where information redundancy permits. Observability analysis (see later) will tell whether any specific parameter can be estimated.

State Variable Equation

The essence of the generalized state estimation approach is to represent the interdependencies of the three main data types within the central state variable equation (2), its solution, and subsequent bad data analysis. This is achieved by augmenting the conventional states and measurements with new state variables and pseudo-measurements to model switch statuses and impedances.

Table 2 summarizes the presently handled measurements as functions of the model states. It roughly illustrates the structure of the Jacobian matrix H of eq.(2). States and measurements may or may not be paired.

Although a WLS approach was selected here, any state estimator method can in principle be used on the re-modeled problem.

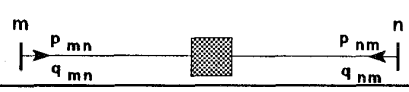
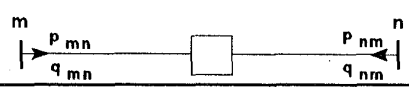
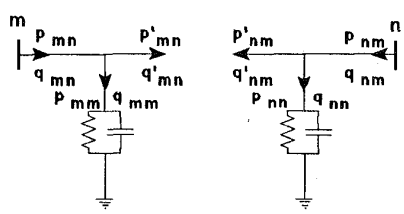
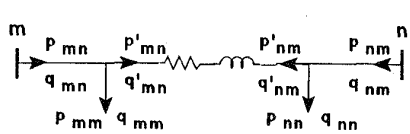
Type	Diagram	Model	Pseudo-Measurement	State
Closed Switch		$p_{mn} = -p_{nm}$ $q_{mn} = -q_{nm}$	Zero Voltage Difference $0 = V_m - V_n$ $0 = \theta_m - \theta_n$	p_{mn} q_{mn}
Open Switch		$p_{mn} = -p_{nm}$ $q_{mn} = -q_{nm}$	Zero Flow $0 = p_{mn}$ $0 = q_{mn}$	p_{mn} q_{mn}
Series Branch		$p_{mn} = p_{mm} + p'_{mn}$ $q_{mn} = q_{mm} + q'_{mn}$ $p_{nm} = p_{nn} + p'_{nm}$ $q_{nm} = q_{nn} + q'_{nm}$	Zero Current Difference $0 = p'_{mn} \cdot V_n + (p'_{nm} \cdot \cos\theta_{mn} - q'_{nm} \cdot \sin\theta_{mn}) \cdot V_m$ $0 = q'_{mn} \cdot V_n + (p'_{nm} \cdot \sin\theta_{mn} + q'_{nm} \cdot \cos\theta_{mn}) \cdot V_m$	p'_{mn} q'_{mn} p'_{nm} q'_{nm}
Shunt Branch		$p_{mn} = p_{mm} + p'_{mn}$ $q_{mn} = q_{mm} + q'_{mn}$ $p_{nm} = p_{nn} + p'_{nm}$ $q_{nm} = q_{nn} + q'_{nm}$	Zero Admittance Difference $0 = p_{nn} \cdot V_m^2 - p_{mm} \cdot V_n^2$ $0 = q_{nn} \cdot V_m^2 - q_{mm} \cdot V_n^2$	p_{mm} q_{mm} p_{nn} q_{nn}

Table 1: Modeling of Switches and Uncertain Branch Parameters

Measurement/ pseudo-measurement	State variables in measurement equation		
	voltage V, θ	turns ratio T, ϕ	flow P, Q
branch flow, target flow (P,Q,I)	X	X	
injection, VAr limit (P,Q,I)	X	X	X
voltage, target voltage (V,θ)	X		
turns ratio, tap limit (T,ϕ)		X	
voltage difference (V,θ)	X		
branch group flow (P,Q)	X	X	X
flow in switch or zero impedance branch (P,Q,I)	X		X
unknown impedance flow (P,Q,I), current difference (I), admittance difference (G,B)	X	X	X

Table 2: Measurements versus States

LOCALIZED MODELING AND SOLUTION

Bad data analysis is a major part of a state estimation solution. This is particularly true when:

(a) modeling at physical level, due to the relatively large number of bus sections and switches, and

(b) using time consuming bad data analysis methods, such as extensive iteration between calculation of normalized residuals and state estimation solutions, or combinatorial analysis.

Fortunately, however, gross errors propagate poorly in state estimation. Therefore bad data localization methods become more and more attractive as the size of the power system increases.

The present generalized approach exploits bad data localization in two ways: it models at physical level only in limited windows, and it solves only designated pockets of the network during bad data analysis. The windows and pockets can change automatically in any ways on the fly, and can overlap and merge: this requires totally flexible data structures and logic, and the ability to track these changes during a solution.

To describe the overall methodology, the entities are introduced:

Region of Interest: the parts of the system eligible for observability analysis, and whose observable islands are subject to bad data analysis. Typically, this is the internal system. The user can redefine this region as desired, for instance during estimator commissioning and troubleshooting.

Bad Data Pocket: an automatically defined scissor cut piece of the region of interest (normally but not always comprised of contiguous devices) that is electrically local to the suspected bad data.

Zoomed Window: a piece of the network whose bus sections and switches are modeled and solved explicitly. Any such window can be defined: (a) by the user permanently or on-the-fly, usually during commissioning or troubleshooting, or (b) automatically when the bad data processor suspects topology errors.

Bus Section Group: any group of substation devices that form an electrical bus when all switches are closed.

Tracking Aspects

With the generalized model, tracking between different modeling levels when processing a single estimation snapshot is essential. Efficiency also dictates (a) the use of previous state estimation information over successive time intervals, and (b) tracking of bus information and topology changes. In the present implementation, (b) is achieved via a scheme that retains the previously assigned bus numbers: new numbers are given only to newly formed buses, and the numbers of vanishing buses are deleted [8].

TOPOLOGY PROCESSING

Extended Islands

The topology processor performs the usual connectivity functions, and defines a network model comprising any mixture of bus-oriented and physical representations. Note that the estimator data base itself can be mixed. For instance, any parts of the external model can be described in power flow form without dummy switches.

The topology processor of course identifies energized, de-energized and grounded electrical islands. However, in this respect it is highly unconventional. Explicitly modeled, nominally "open" (or unknown) switches form connections to "electrical island extensions". It is not possible to determine the energization statuses and connectivities of such islands until after the state estimation solution is obtained, and the bad data analysis on the explicitly modeled switch statuses is performed. Consequently, the central estimation solution and bad data analysis must be capable of handling network models comprising "extended electrical islands", that is, conventional islands plus the extensions.

Plausibility Tests

Topology processing as implemented here includes a wide range of basic plausibility tests. Additional more complicated tests could be added [9]. But sophisticated checks associated with network flow and voltage consistency are no longer necessary, since they are inherently handled by local state estimation and bad data analysis at bus-section/switch level. The plausibility tests as currently implemented upgrade the confidence ratings of good measurements, including switch statuses.

Local Estimation for Topology Errors

The topology processor automatically "zooms" the model surrounding each suspect bus section group, that is, a group containing switches: (a) whose statuses have changed since the last time, or (b) with poor confidence ratings.

The generalized estimator then *performs local observability, state estimation and bad data analysis at physical model level.* When necessary for observability, the measurements from the far end of a connected branch are utilized in this model.

The purpose of these solutions is to verify and flag consistent measurements and statuses before the main state estimation, as far as possible. As a result, the confidence ratings of both measurements and switches are updated for use during the subsequent bad data analysis and the next execution of the topology processor.

OBSERVABILITY ANALYSIS

A state variable is observable if it can be estimated. Classical observability analysis is "topological". It makes a yes/no pronouncement on the observability of each state variable. The best logic-based versions are very complicated, and need expert hard-coded modifications whenever new or altered power system models, measurements and state variables are introduced.

With the present new measurement and state variable types, and the prospect of other types in future, a more general, flexible, approach was needed. Very efficient combinations of observability methods are possible [10]. The approach adopted here is:

Stage 1: A simple topological observability process reliably and very rapidly finds all easily-flagged observable states.

Stage 2: The "numerical-topological" method of Refs. 11-12 is now applied only to those small network portions remaining unanalyzed after Stage 1.

Both stages search for the maximum observable islands, less conservatively than algorithms requiring pairing. The result is a topological observability method with great speed and generality. It exactly mimics the central state estimation solution and code itself. That is, the observability software accommodates the same measurements, states and models as the central state estimation method, and is guaranteed to be non-obsolescent whenever these are extended.

The present implementation of the observability analysis can utilize unpaired measurements. As a consequence, no fictitious missing measurements need to be provided. Pairing of measurements is automatically taken care of where possible.

Because the electrical islands defined by the topology processor are "extended", so too are the observable islands defined by the observability analysis. Within a single extended observable island, it becomes necessary to assign a separate angle reference to each sub-island that one or more estimated switches can potentially isolate. Like electrical islands, observable islands are only fully resolved at the end of the entire estimation process.

CENTRAL ESTIMATION SOLUTION

The WLS approach, thought to have fewer weaknesses than more recent alternatives, was retained in this implementation. A coupled algorithm is used, efficiently exploiting measurement and state pairing where available, in conjunction with a "mixed-block" double precision version of the Fast Givens orthogonal solution [13-15]. This provides outstanding immunity to ill-conditioning: the high confidence pseudo-measurements required by the generalized estimation approach can be handled stably using weightings as large as desired. Explicit equality constraint imposition is unnecessary.

No topological observability process guarantees problem solvability. It is still possible for an "observable" state variable to acquire such a low numerical sensitivity to the rest of the states that it cannot be estimated in the central solution. In the present orthogonal factorization, this is guarded against using the "discardable measurements" approach [17], modified to accommodate the new flow pseudo-measurements for switches and parameters.

The solution engine must include safeguards for other conditions associated with matrix singularity: (a) the voltages in de-energized portions of the network are not allowed to fall below a specified threshold, (b) when zero-impedance loops have unobservable flows, the flow division is automatically assigned using the switch ratings.

There are many other aspects of the overall implementation, some of which will be briefly mentioned in this paragraph, as well as in Appendix I. Locally acting controls such as LTC transformers and phase shifters have been modeled via additional pseudo-measurements and designated state variables. Generator MVAr limits, load MW and MVAr limits, and bus/bus-section voltage limits are enforced, with back-off logic. The discretized and limited state variables are handled by introducing heavily weighted measurements corresponding to the values at which these variables are to be fixed. Limiting and discretization are very conveniently and efficiently achieved by partial orthogonalization, to add the new measurements to the already available matrix factors.

BAD DATA PROCESSING

The Pocketing and Zooming Processes

The effects of bad data in a power system normally only manifest themselves locally. This is exploited by conducting the time-consuming part of bad data analysis only on network pockets [18]. Each pocket is automatically defined. It can be modeled at bus/branch level, zoomed level, or any mixture of the two. Whenever zooming is present, the estimation also solves for the switch flows. Redundancy permitting, an incorrect status is identified as a gross error in the switch pseudo-measurement (Table 1).

The basic steps of the localized bad data processing are:

A. Perform topology processing, observability analysis and the central state estimation solution in the region of interest.

B. Perform bad data screening on the entire observable network, using the simple largest weighted residuals approach to identify the initial suspect measurement set (see Note 1 below). Exit if this set is null.

C. Use the measurement/state adjacency algorithm of Appendix II to build a bad data pocket around each suspect measurement location. Each such pocket is a separate observable island extracted from the entire network. No separate observability analysis is performed since the original network was initially observable (see Note 2 below).

D. Perform state estimation on the bad data pockets, and use the largest normalized residuals method to identify their bad data (see Note 1 below).

E. Test the pocket for suspect topology errors, based on poor confidence ratings (see Note 3 below) and/or adjacent bad measurements. If the test is positive, redefine the pocket to include zoomed modeling:

   (i) Apply the algorithm of Appendix II to the already available Jacobian matrix structure of the existing pocket, to define the zoomed windows surrounding the suspect bad data.

   (ii) With the zoomed windows explicitly modeled, perform network topology processing on the entire network.

   (iii) Re-apply the algorithm of Appendix II, now to the entire network's Jacobian matrix structure, to obtain the pockets containing the zoomed windows (see Note 4 below).

   (iv) Repeat the calculations of step D.

F. If necessary, restore the network comprising the region of interest. Eliminate identified analog bad data. Change the statuses of identified bad switches. Go to step A.

Note 1: In the present implementation, weighted and normalized residuals are used for bad data screening and analysis respectively. However, this multi-level approach can easily be extended to use other methods. For instance, normalized residuals and combinatorial analysis [19] may be utilized for bad data screening and analysis respectively.

Note 2: Observability can sometimes be lost with pocket boundary buses. It is restored automatically by adding a pseudo-measurement of a state, obtained from the original non-pocketed solution.

Note 3: Each measurement has a "confidence rating", varying from 0 (not suspect) to 5 (highly suspect), which is inherited from previous estimations and is updated by the program (or manually as desired).

Note 4: The pockets after step E(iii) may be quite different from those before step E(i). For instance, due to the explicit modeling of open switches, observable but de-energized parts of the network may now become included within the pockets. As in Note 2, no separate observability analysis is performed for these pockets. Observability is restored by pseudo-measurements of voltage and flow states.

Good Data Analysis

In bad data analysis, it is equally important to identify measurements that had wrongly been classified as bad data, or were previously identified as bad data but have become good over the course of time. "Measurement error analysis" techniques are utilized for this purpose [18]. When good measurements are identified, they are re-instated into the valid measurement set, and the state estimation solution and bad data analysis are repeated.

The Bad Data Analyzer

The central bad data analyzer is based on the normalized residuals approach using bad data removal. With the orthogonal method, this is analogous to reintroducing these measurements with negative weights, updating the matrix factors using very efficient "partial orthogonalization" techniques [5,20].

Starting with an initial solution, every time bad (good) data is identified within the trusted (suspected) measurement set, the effects of its removal (insertion) on the estimated states, residuals, and the residual covariance matrix elements must be correctly re-calculated before the next identification step [18]. In the present implementation, there is a user option to calculate states and residuals in two modes: (a) the "linear mode" in which linear updating formulas are utilized [21], or (b) the "non-linear mode" where a complete non-linear state estimation solution is performed to calculate them. In both cases the residual covariance matrix is modified using linear updating formulas. While mode (a) is computationally more efficient, mode (b) works better when the accuracy of the linear model is not satisfactory, such as when large numbers or magnitudes of bad data are present.

PARAMETER ESTIMATION

Parameter estimation is still an emerging field [22-24]: development continues on the effective computation of network parameters and how to implement the results in an EMS environment. The approach adopted here has some significant distinguishing features. One such distinction arises from the fact that a suspect impedance parameter is omitted completely from the estimation process. This eliminates the problem in existing methods that a very bad parameter could jeopardize the entire estimation process.

In this approach, a parameter is estimated in each of a sequence of time snapshots (during which system load and topology can change arbitrarily), independent of previous estimates. Each time, its confidence limits are calculated via the state covariance matrix. ***Then the parameter's statistical variation over the sequence is analyzed to produce its composite estimate and confidence limit.*** This approach is consciously conservative. An update to the data base becomes sanctioned if and only if the composite estimate has high confidence. This contrasts with substantially more aggressive approaches, where the estimate of each parameter (and the entire system state) at a snapshot is a function of past estimates, and is vulnerable to "estimate drift": progressive corruption of the parameter estimates under the influence of various sources of inaccuracy.

In daily operation, static parameters such as branch resistances and reactances, and fixed transformer taps will need to be estimated only on demand, over a finite number of estimation cycles. On the other hand, dynamic parameters such as LTC transformer taps always need to be estimated, since errors in them can degrade the current estimation results significantly.

The generalized approach prescribes that estimated parameters (*i.e.* their pseudo-measurements) be treated the same way as analog and switch measurements during the estimation solution. It is only by simultaneously analyzing the interactions of the analog measurements, parameters, and switch statuses that the statistically best bad-data candidates can be determined.

ILLUSTRATIONS

This section briefly illustrates the generalized approach, implemented as production software, and applied to an observable system with 2046 buses, 2632 branches and 7716 telemetered measurements. Random errors are generated to simulate measurement noise on the exact power flow solution. Bad data analysis uses an identification threshold of $\pm 3\sigma$.

Bad Data Processing with Pocketing and Zooming

The example here has two gross errors in different locations: Error 1 is on an analog measurement, and Error 2 is a closed switch wrongly metered as open. The switch error creates an erroneous electrical island, comprising the upper part of Figure 1. All the following processes are completely automatic.

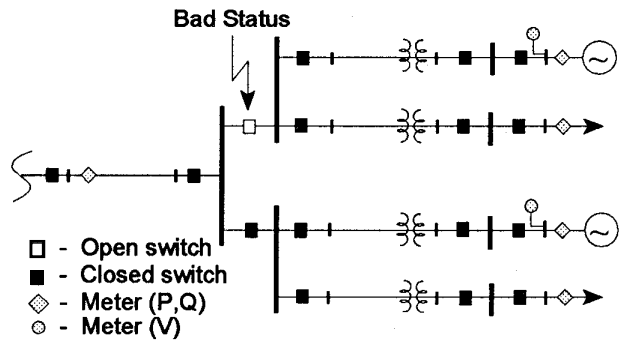


Figure 1: Switch Error and Erroneous Island

The initial state estimation solution starts from a flat voltage profile and takes 9 iterations. At the solution point, bad data screening identifies 5 suspect measurements. Three pockets of 53, 25 and 3 buses respectively are created (one of which is the erroneous island). Bad data analysis on these pockets now identifies Error 1 as the only bad measurement in Pocket 1. However, it finds two analog bad data points each in Pockets 2 and 3, triggering switch error analysis.

Zoomed windows, modeling all the switches explicitly, are then created around the suspected measurements in Pockets 2 and 3, and they merge into the network portion shown in Figure 1. A new pocket containing 60 nodes and 38 switches is formed around this window. Combined measurement and status bad data analysis now identifies the erroneous switch as closed. Its status is corrected, and the topology processing, state estimation and bad data analysis are repeated using the full system model. At this point, no further bad data is identified.

On a 200MHz Pentium Pro PC, the entire process above takes 3.4 seconds. About one third of the time is for the initial state estimation (before bad data processing). For comparison, with no pocketing, the same solution takes 50% longer. However, this difference will increase considerably as the overall size of the system increases, and as more sophisticated bad data analysis methods are introduced, because the pocket sizes themselves do not increase.

The software also includes the optional "hybrid" orthogonal solver [16], which runs the same case 10-15% faster, with negligible accuracy loss. A still-faster decoupled version is planned.

Estimation of Impedances

Using the same power system, the series impedances of three branches are corrupted by multiplying them by 2, 1.5 and 0.5 respectively. In addition, a P,Q flow measurement with bad data is simulated on the last of these branches, posing the obvious risk of error smearing between it and the estimated parameter (Figure 2).

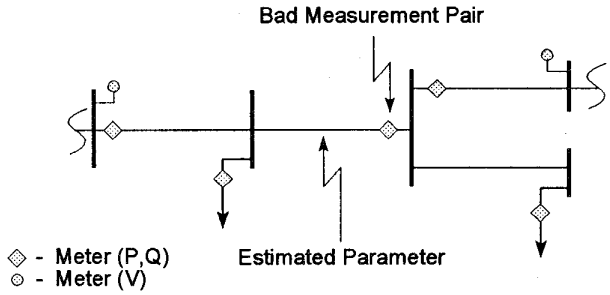


Figure 2: Parameter Error Identification

An initial "ordinary" state estimation solution with no parameter estimation identifies a total of 14 bad analog measurements, in clusters of 3, 7, and 4 near to the respective corrupted branches. Now the solution is repeated, this time including parameter estimation. Nine branches in the three cluster vicinities are nominated as having suspect parameters, their impedances become removed from the problem (made dormant), and their parameters become estimated. Table 3 shows the results for the 3 relevant branches. At this point, bad data analysis pronounces the absence of gross errors.

Branch Type	Original		Bad Data		Estimated	
	R	X	Mult.	Ident.	R	X
Line	0.0003	0.0011	2.0	3	0.0004	0.0013
Xfmr	0.0006	0.0266	1.5	7	0.0005	0.0265
Line	0.0025	0.0156	0.5	4	0.0029	0.0153

Table 3: Parameter Estimation Example

CONCLUSIONS

This paper describes a generalized treatment for the three main classes of data in static power system state estimation. It gives many design details for the generalized approach, and for the overall estimation process within which it was implemented. Bad data handling is emphasized throughout.

The state estimation field will continue to evolve in various directions. However, it is thought that the generalization principle is fundamental, and is likely to become widely adopted in other estimators in future. It has no strong association with any particular central state estimation approach or algorithm.

The above applies equally to the concepts of "pocketing" and "zooming". With the prospect of regional-sized state estimation models, there is considerable incentive to localize bad data analysis, because the bad data pockets and windows do not grow in size with the total number of buses. Other pocketing algorithms are possible.

Switch status error handling is clearly more precise when the relevant switches are not eliminated from the network model. If the advantages of physical level modeling become accepted, increased use will be made of breaker flow metering in future.

Parameter estimation is an area of considerable importance both for data base debugging and for tracking parameter changes over time, whether slowly or (as in the case of corona losses) quite rapidly. The present parameter estimation based on statistical averaging of a sequence of parameter snapshots seems to be a good low-risk approach, but others are possible. High measurement quality and redundancy will of course continue to be the main ingredients of successful parameter estimation.

ACKNOWLEDGEMENTS

Much of the reported development was undertaken as a joint project between PCA and CAE Electronics from 1992-95. We specially wish to acknowledge Dr. Hadi Banakar (CAE) for his important help and advice, both technically and managerially, throughout the project. We would also like to thank Mauro Prais (PCA) and Tim Ratzlaff (CAE) for their valuable contributions during the project.

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APPENDIX I
Internal/External Solutions

The power system comprises an "internal" and an "external" network, the latter of which can of course have redundant measurements [25]. The system is solved in a two-part process. In Part 1, the extended observable islands in the internal system are defined. Estimation and bad data processing are performed on them. Any non-observable parts are ignored, and are treated as external.

When Part 2 is required, a Bus Scheduler function must provide the predicted external data. Interchange schedules are used where available. Then the external system is attached to the already-solved internal network, and the whole system state is estimated, protecting the observable portion solution with very high weightings. Differentiated external weightings are used to promote realistic tie-line flows, and to distribute boundary mismatches into the external network.

Uncertain data in the external network can affect state estimation convergence. If any bus or bus section voltage collapses, it is propped up with a voltage pseudo-measurement and the anomaly is reported. Thus, a convergent external model is virtually assured.

APPENDIX II
Measurement/State Adjacency Algorithm

The algorithm used to determine the pockets or windows starts from an initial suspect measurement set. It automatically merges overlapping pockets or windows. Its basic steps are as follows:

(a) EXP = 0.
(b) Using the Jacobian matrix structure, select the state variables "adjacent" to the suspect measurement set (state expansion). Go to step (d) if EXP = MXEXP.
(c) EXP = EXP + 1. Using the Jacobian matrix structure, update the suspect measurement set by adding those measurements which are functions of the selected state variables (measurement expansion). Go to step (b).
(d) Define the sub-networks (pockets or windows) formed by the final suspect set of measurements and states.

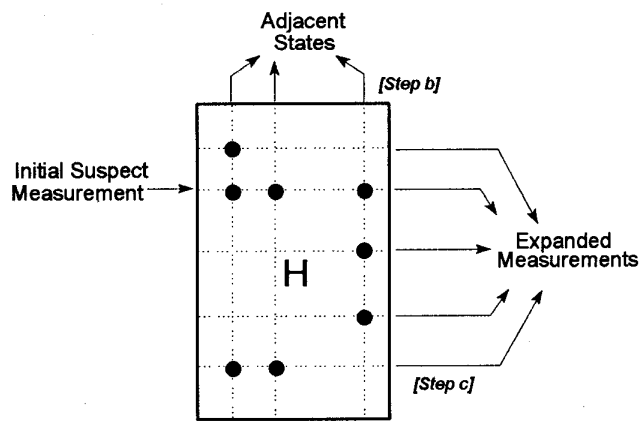

Figure 3: Illustrating Measurement/State Adjacency Algorithm

Figure 3 shows the basic mechanism of the algorithm, utilized both for the determination of bad data pockets and zoomed windows. The size of the pocket or window depends on the number of state/measurement expansions around the suspect set of measurements. This is governed by parameter MXEXP, which is different for pockets and windows, and for maximum efficiency and accuracy needs to be determined for a given system.

BIOGRAPHIES

Ongun Alsaç is vice president of Power Computer Applications (PCA), which he co-founded in 1984. He received a Ph.D. from UMIST, UK and is a Fellow of the IEEE. He has served on each PICA Technical Committee since 1987.

Narasimham Vempati has been a senior engineer at PCA since 1993. He was previously at CDC-Empros, and co-authored the state estimation paper that won the 1993 W.R.G. Baker award for the best IEEE paper. He received a Ph.D. from U. of Texas (Arlington).

Brian Stott is president of PCA, which he co-founded in 1984. He received a Ph.D. from UMIST, UK and is a Fellow of the IEEE.

Alcir Monticelli is professor of electrical engineering at the University of Campinas (UNICAMP), Brazil, where he received a Ph.D. He is a Fellow of the IEEE. He has served on each PICA Technical Committee since 1987.